

Advancing Solar PV Component Inspection: Early Defect Detection with UAV Based Thermal Imaging and Machine Learning

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Abstract — The timely detection of defects in electrical components of solar systems is of utmost importance to prevent potential catastrophic failures. This paper proposes a non-intrusive imaging system that utilizes ultraviolet technology to monitor the condition of components in small distributed generation plants. The system is designed to be mounted thermal camera on an unmanned aerial vehicle (UAV), enabling thermal imaging of solar panels. Subsequently, the captured images undergo various image processing techniques to extract relevant geometric and statistical features. These processed data are then analyzed using a machine learning classifier algorithm, aiming to identify areas with defects. Multiple algorithms are employed and compared to enhance and assess the accuracy of defect identification. This innovative approach, combining UAV-based thermal imaging and machine learning, presents a promising solution for early defect detection in solar PV components. It has the potential to significantly improve maintenance practices and mitigate the risks of system failures.

I. INTRODUCTION

With the growing adoption of solar photovoltaic (PV) systems as a clean energy source, ensuring their efficient and reliable operation is paramount. Detecting defects in PV components at an early stage is crucial for preventing system inefficiencies and potential failures. However, traditional manual inspections and visual assessments have limitations in terms of subjectivity, time consumption, and the ability to identify hidden or early-stage defects. Moreover, as solar PV installations continue to expand, manual inspections become impractical. To overcome these challenges, researchers have turned to advanced technologies to enhance defect detection in solar PV components. Unmanned aerial vehicles (UAVs), equipped with high-resolution cameras and thermal imaging sensors, offer a valuable solution for conducting aerial inspections of solar PV installations. Thermal imaging, in particular, has proven effective in detecting temperature variations that can indicate underlying defects like cracks or faulty connections. Analyzing thermal images enables non-intrusive defect identification and timely maintenance interventions. Machine learning algorithms have also played a significant role in improving defect detection. By training models on large datasets of thermal images and corresponding defect labels, machine learning algorithms can learn patterns and correlations to accurately classify thermal images and identify defective areas.

The integration of UAV-based thermal imaging and machine learning presents a promising approach for early defect detection in solar PV components. It offers efficient, comprehensive, and non-intrusive inspections, reducing dependence on manual labor and improving defect identification accuracy. This combination has the potential to transform the inspection of solar PV systems, leading to improved maintenance practices and enhanced system reliability.

In this study, we propose a novel system that utilizes UAV-based thermal imaging and machine learning techniques for early defect detection in solar PV components. We describe the hardware setup, image processing techniques, and machine learning algorithms employed. Through experimental analysis, we evaluate the performance and accuracy of different algorithms in identifying defects. The research aims to advance defect detection methods in solar PV systems and contribute to the development of reliable renewable energy infrastructure.

II. METHODOLOGY

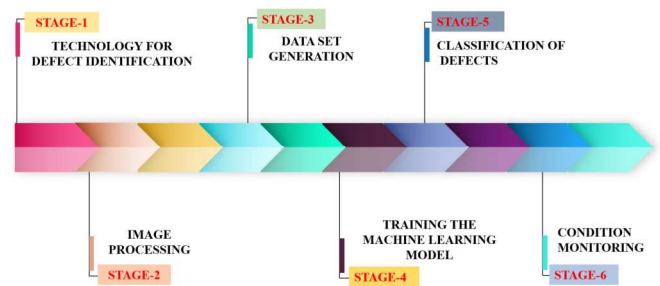


Fig. 1. The methodology of the proposed system.

A. Technology for defect identification

Infrared thermography emerges as a non-destructive technique for detecting and anticipating potential faults in various electrical assets by identifying temperature abnormalities. A drone fitted with thermal imaging camera used to scan and capture the solar modules for their defects. The faulty regions identified as hotspot in the captured images.

B. The image processing

Analyzing the resulting images to distinguish between healthy and faulty components based on the presence of hotspots in the solar panels. These faulty components were identified by examining the temperature readings from the images. This processing workflow enabled the extraction of relevant features and enhanced the visual representation of potential defects in the solar module images, facilitating further analysis and defect identification.

C. The dataset generation

This is achieved by employing an image region analyzer, which generates a table containing region properties. Each row in the table corresponds to a region identified in the image, while each column represents a specific property of that region, such as area, perimeter, and orientation.

D. Training and Machine learning classification

To develop algorithms that possess the capability to autonomously gather and learn from data, detect patterns, and derive informed decisions based on the available information. Various classification algorithms are applied to process this dataset, with the output response being the type of fault. A 5-fold cross-validation scheme is employed for validation, Within each fold, 70% of the data is used for training, while the remaining 30% is employed for testing the algorithms' performance.

E. The classification prediction

The dataset consisting of geometrical and statistical features is employed to train and test different SVM algorithms. The performance evaluation of the model is based on the highest percentage of validation accuracy and test accuracy. Furthermore, the evolutionary model will be employed to make predictions on new datasets.

F. The classification prediction

The predictions derived from fault classification can be leveraged for predictive maintenance procedures, enabling the confirmation and validation of the specific faults present in the equipment. At this stage is to collect information about the equipment's condition and gain a deeper understanding of the causes of equipment failures and patterns of equipment degradation.

III. EXPERIMENTATION AND INTERPRETATION OF RESULTS

The drone equipped with a thermal imaging camera to capture thermal images of a solar power generation system with 12 panels connected in series parallel connection under varying environmental conditions. The thermal images were then processed using a proposed system to identify and classify faults into three categories: less faulty (class 0), medium faulty (class 1), and severe faulty (class 2) based on the difference in the temperature of hotspot region and ambient.

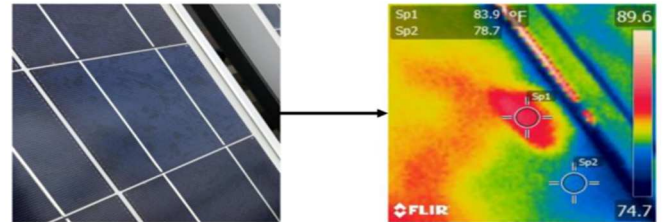


Fig. 2. Capturing the thermal images through drone.

A dataset was created for the machine learning algorithm using images from various reference papers, which were processed to obtain the geometrical features of the faulty regions through image region analysis. The resulting 350X12 dataset was used for training and testing the algorithm, and a prediction dataset was formed with 12 images with small defects. The images were segregated based on the temperature of the defected area in the solar module due to fault as hotspot and the ambient temperature of the sarroundings. The criteria of labelling is done as per the Table I.

Fig. 3. Shaded fault of module captured asthermal images

TABLE I
SUMMARY OF LABELLING IMAGES USED FOR ANALYSIS



DIFFERENCE IN TEMPERATURE	CLASSIFICATION OF IMAGE
$\Delta T < 10^{\circ}\text{C}$	TYPE 0
$\Delta T = (10-20)^{\circ}\text{C}$	TYPE 1
$\Delta T > 20^{\circ}\text{C}$	TYPE 2

Thus captured images were processed through image processing techniques to find the faulty region. The initial stage of it convets the RGB image into grey scale ine to identify the clear faulty region. Thus through image region analyser, the geometrical and statistical image features are extracted to know the exact faulty region over the surface of module as defected area due to fault. The data set obtained through processing different images mentioned in the table II. The training and testing data set is formed with 300 images labelled as per table I. with 12 different image geometrical and statistical features. The size of training data set is formed 350X12. A 23X12 prediction data set is formed for new callsification of faults for the images captured through drone. These data set were processed through various machine learning classifier algorithms.

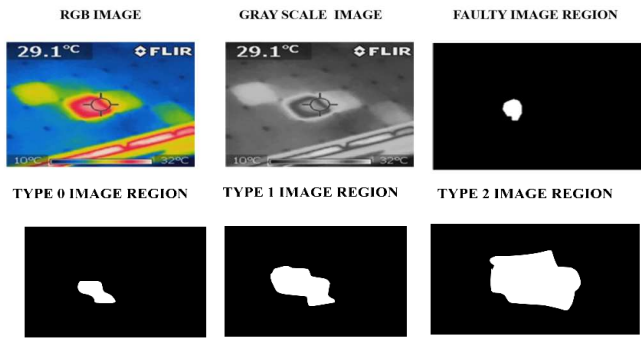


Fig. 4. Different stages of Image Processing

TABLE II
SUMMARY OF IMAGES USED FOR ANALYSIS

CLASS	TYPE 0	TYPE 2	TYPE 3	TOTAL
TRAINING	84	120	146	350
PREDICTION	3	8	12	23

10 different types of Machine Learning classifiers with 31 different classifier models are used in the classification learner app in MATLAB 2023a (Online) version for the analysis of fault classification of solar module. The model performance is assed through the validation and testing accuracies of the models. The classifier models with highest validation, testing accuracies are evaluated as model to predict the classification of faults for the data set of images captured through drone. Out of 10 classification algorithms, Support vector machine models shows the better accuracies. The Different SVM-based machine learning algorithms were used to process the dataset. The information obtained from this process can be used to implement predictive maintenance procedures by verifying and identifying the type of fault in the solar power generation system. The accuracies of the different SVM models are shown if the fig.5.

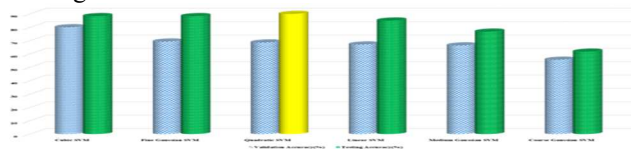


Fig. 5. SVM model performance.

IV. CITING PREVIOUS WORK

As per IEA reports, the rise of solar installations and their ratings as on date are expected to be exponentially [1-2]. The conventional inspection methods are time consuming in detecting the defects of panels. Researchers are trying to implement advanced methods over the sensor-based detection. The image-based approaches are exhibiting better performance in defect identification. The identification of faults identified through color images for thermography study gives novel approaches [2-5]. The image processing methods implemented to identify faulty regions are accurate to depict the damage caused by various faults, giving an initiation to detect the features of faulty areas [6-7]. The fault classification based on the data approaches is an accurate approach using machine learning [8-10].

V. REFERENCES

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